

Vadose Zone

Katie Markovich, EPS 522/ ENVS 423L (Fall 2025)

Housekeeping

- Streamflow problem set (#5) due Tuesday night
- Lit. review due tonight
- Guest lecture next class

Learning Objectives

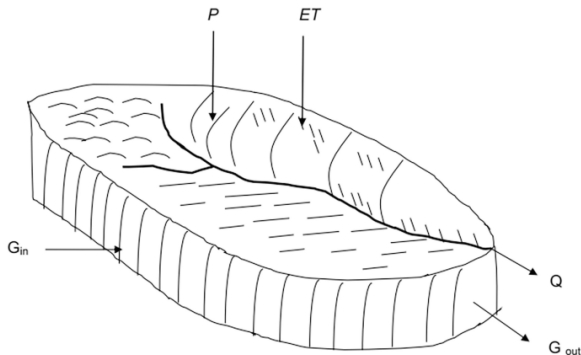
- 1 Material properties of the subsurface
- 2 Water content and pressure
- 3 Infiltration
- 4 Redistribution

Regional Water Balance

$$\hat{P} + G_{in} - (Q + ET + G_{out}) = \Delta S$$

The vadose/unsaturated zone governs the partitioning of P to ET , Q , and groundwater recharge.

"Hydrology is mostly soil physics" (Youngs, 1988)

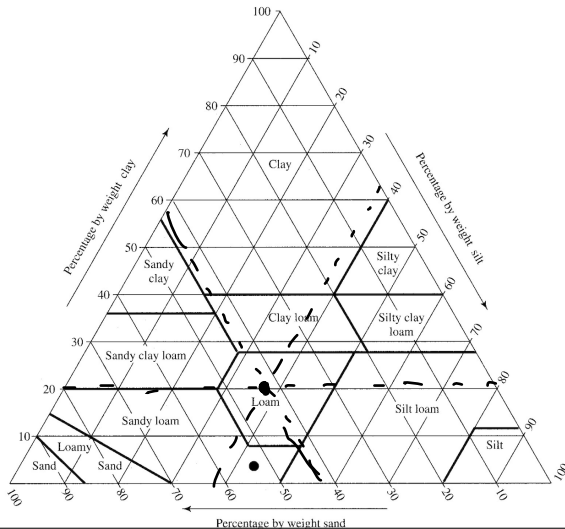


Where:

- P = Precipitation
- ET = Evapotranspiration
- Q = Stream outflow
- G_{in} = Ground water in
- G_{out} = Ground water out

Soil Texture Ternary Diagram

What type of soil is 43% sand, 20% clay, and 37% silt?



Notes on Loam

Mostly silt and sand with a lesser amount of clay (40-40-20).

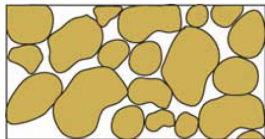
Well-drained soil with good water holding capacity, nutrients/organic material, and is easier to till than soils with higher clay contents. Ideal for farming!



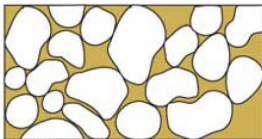
Define that greek letter!

- θ theta, volumetric water content (soil moisture) $\frac{V_w}{V_a + V_w} \frac{L^3}{L^3}$, ranges from 0 - ϕ
- ϕ phi, porosity, $\frac{V_a + V_w}{V_{pm}} \frac{L^3}{L^3}$, 0 - 50%
- ψ psi, pressure/tension head $P/\gamma_w = \psi$ matric potential/suction ψ - 100 cm
units of L ψ - 1000 cm

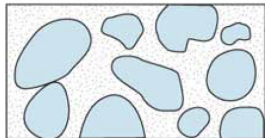
Which has low/high porosity?



Gravel
well sorted, 25-50%



Gravel - Sand - Clay
poorly sorted, 20-35%



Cemented Sandstone
10-20%



Clay ~ 5%



Limestone 7-5%



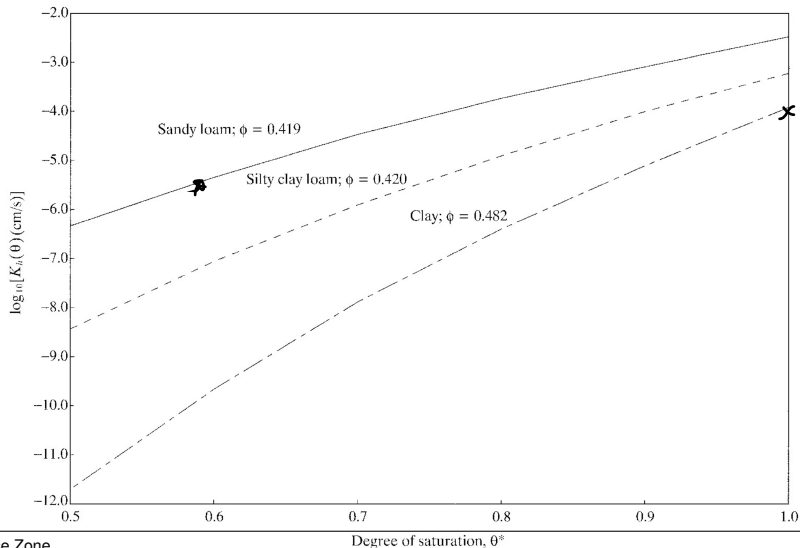
Shale 1-1%

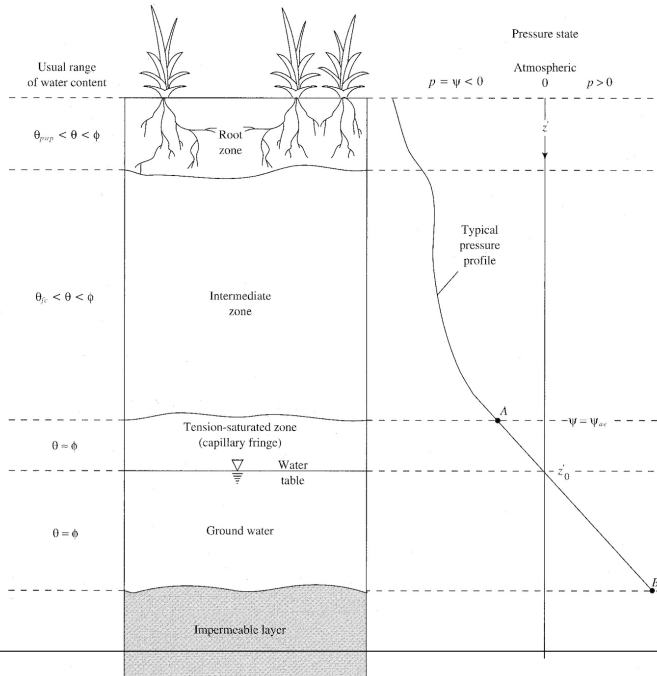
Which has a higher K?

Completely saturated clay or 60% saturated sandy loam?

$$S = \frac{\theta}{\phi}$$

K^*





Putting it all together

Even 1-D vertical flow in the unsaturated zone is complex and dependent on the properties (K , ϕ) and state (θ , ψ) of the porous medium.

$$q_z = -K(\theta) \cdot \left[1 + \frac{d\psi(\theta)}{dz} \right]$$

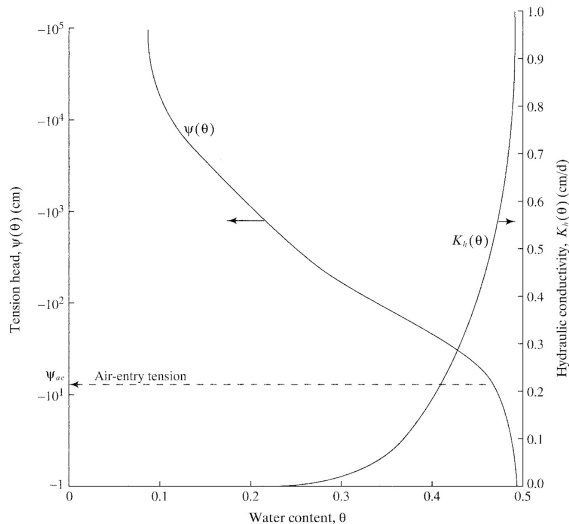
Analytic Approximations of $\psi(\theta)$ and $K(\theta)$

$$|\psi(\theta)| = |\psi_{ae}| \cdot \left(\frac{\phi}{\theta}\right)^b$$

where b is the **pore-size distribution index**.
This only applies to $|\psi| \geq |\psi_{ae}|$.

$$K(\theta) = K^* \cdot \left(\frac{\theta}{\phi}\right)^c$$

where K^* is saturated K and c is the **pore-disconnectedness index**.



Pore-size Distribution

Soil Texture	ϕ	K_h^* (cm s ⁻¹)	$ \psi_{ae} $ (cm)	b
Sand	0.395 (0.056)	1.76×10^{-2}	12.1 (14.3)	4.05 (1.78)
Loamy sand	0.410 (0.068)	1.56×10^{-2}	9.0 (12.4)	4.38 (1.47)
Sandy loam	0.435 (0.086)	3.47×10^{-3}	21.8 (31.0)	4.90 (1.75)
Silt loam	0.485 (0.059)	7.20×10^{-4}	78.6 (51.2)	5.30 (1.96)
Loam	0.451 (0.078)	6.95×10^{-4}	47.8 (51.2)	5.39 (1.87)
Sandy clay loam	0.420 (0.059)	6.30×10^{-4}	29.9 (37.8)	7.12 (2.43)
Silty clay loam	0.477 (0.057)	1.70×10^{-4}	35.6 (37.8)	7.75 (2.77)
Clay loam	0.476 (0.053)	2.45×10^{-4}	63.0 (51.0)	8.52 (3.44)
Sandy clay	0.426 (0.057)	2.17×10^{-4}	15.3 (17.3)	10.4 (1.64)
Silty clay	0.492 (0.064)	1.03×10^{-4}	49.0 (62.1)	10.4 (4.45)
Clay	0.482 (0.050)	1.28×10^{-4}	40.5 (39.7)	11.4 (3.70)

Infiltration

Process by which water arriving at the soil surface enters the soil. Some key terms include:

- **infiltration rate**, $f(t)$, (L/T). Rate at which water enters the subsurface.
- **water-input rate**, $w(t)$, (L/T). Rate at which water arrives at the surface (rain, snowmelt, irrigation).
- **infiltration capacity**, $f^*(t)$, (L/T). Maximum rate at which infiltration can occur.
- **depth of ponding**, $H(t)$, (L). Depth of water standing on the surface.

Infiltration Conditions

- 1 No ponding. "Supply controlled"

$$H(t) = 0, f(t) = w(t) \leq f^*(t)$$

- 2 Saturation from above. "Profile controlled"

$$H(t) > 0, f(t) = f^*(t) \leq w(t)$$

Infiltration excess
"Hortonian"

- 3 Saturation from below.

$$H(t) > 0, f(t) = 0$$

Saturation excess
"Dunnean"

Measuring Infiltration in the Field

Ring infiltrometers: direct measurement of infiltration over small area. Add specified volume of water or maintain constant ponding depth during time interval (Δt).



Figure 8.14 A double-ring infiltrometer connected to an apparatus that maintains a constant water level in each ring (photo by author).

Measuring Infiltration in the Field

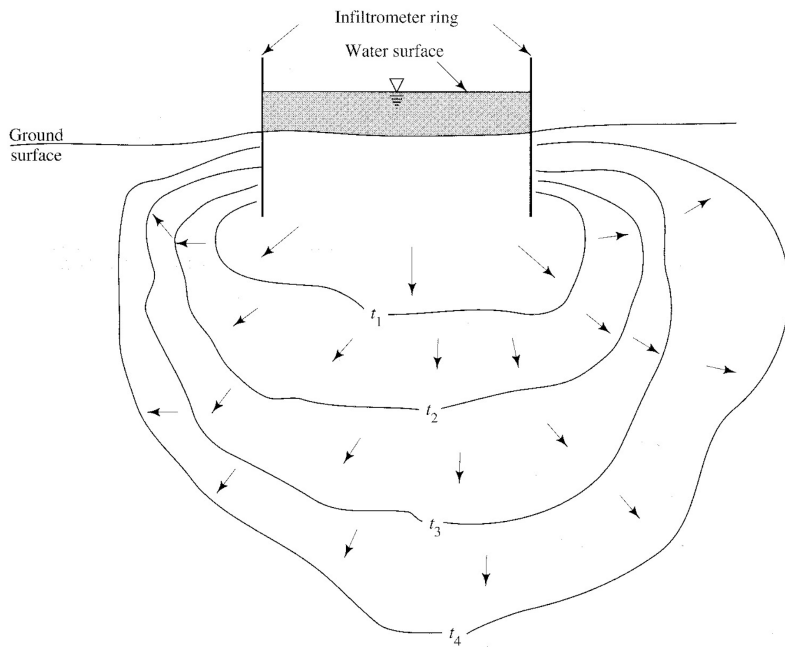
Infiltration is measured by 1) rate of water level decrease inside the ring, 2) rate of additional water needed to maintain constant level, or 3) a water balance equation:

$$f(t) = \frac{W - \Delta H \cdot A}{\Delta t}$$

W is water input, ΔH is change in ponded water level inside ring, and A is cross-sectional area.

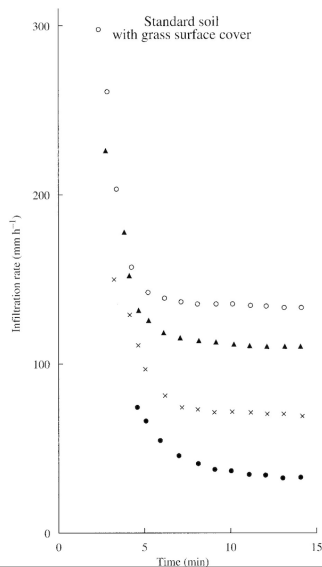


Figure 8.14 A double-ring infiltrometer connected to an apparatus that maintains a constant water level in each ring (photo by author).



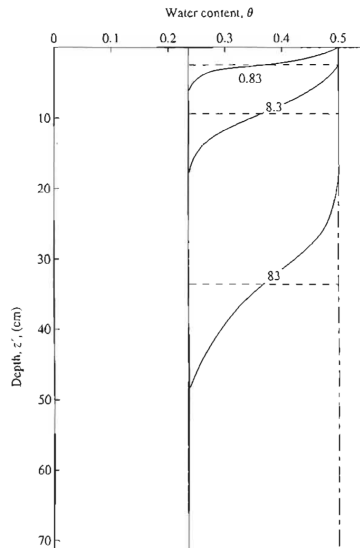
Infiltration Rate

Infiltration studies show that, when there is saturation from above, infiltration rates start out high and then decline rapidly until asymptotically reaching a constant rate equal to saturated hydraulic conductivity.

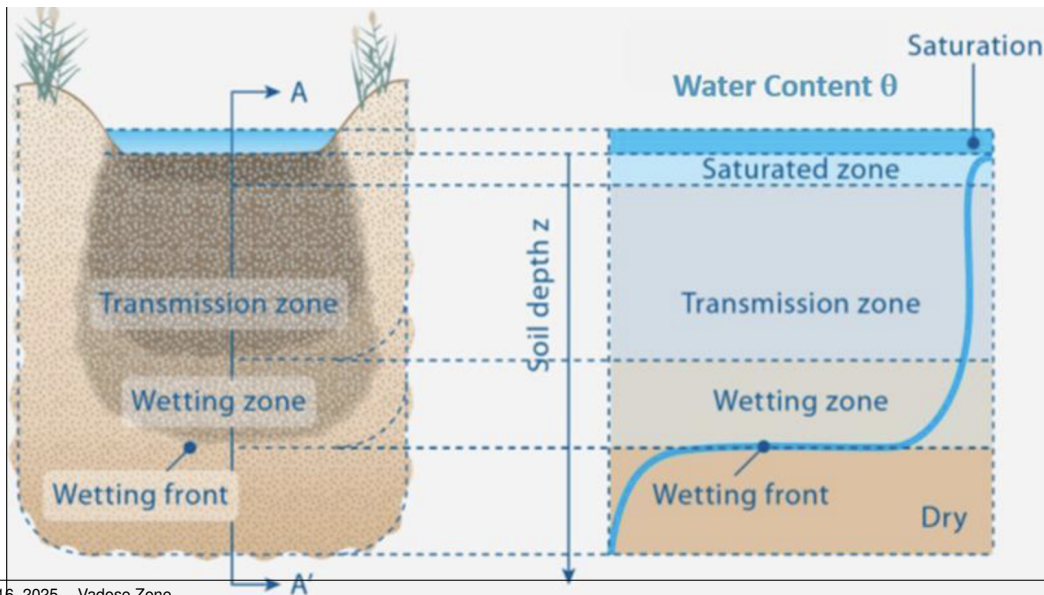


Soil Moisture Profile

Infiltration results in a "wetting front" that moves downward over the course of the infiltration event. The sharpness of this front depends on soil texture, with clays having stronger capillary effects and a diffuse front and sands showing a sharper front.



Soil Moisture Profile



Factors Affecting Infiltration

- water input rate or depth of ponding
- saturated hydraulic conductivity of soil profile
- antecedent wetness of the soil profile
- inclination and roughness of soil profile
- chemical characteristics of soil profile
- physical/chemical characteristics of water

Factors Affecting Infiltration

- water input rate or depth of ponding
 - ▶ higher rate: higher $f(t)$ until $v(t) > f^*(t)$
but it depends
- saturated hydraulic conductivity of soil profile
 - ▶ organic content: higher generally but depends!
 - ▶ frost: lower $f(t)$, freeze/thaw leads to preferential flow pathways
 - ▶ swelling/drying: swelling $\rightarrow$ low $f(t)$, drying leads to cracks, high $f(t)$
 - ▶ sediment washing: lower $f(t)$
 - ▶ human modification: pavement + compaction, lower $f(t)$

Factors Affecting Infiltration

- antecedent wetness of the soil profile
 - ▶ initially wet soil: *medium $f(t)$*
 - ▶ initially dry soil: *high $f(t)$, it depends!*
- inclination and roughness of soil profile
 - ▶ steeper slope: *low $f(t)$*
- chemical characteristics of soil profile
 - ▶ hydrophobic soil: *very low $f(t)$*

Modeling Infiltration

The **Richards' Equation** combines Darcy's Law with conservation of mass to provide a basic theoretical equation for modeling flow in unsaturated porous media.

$$\frac{\partial \theta}{\partial t} = -\frac{\partial K(\theta)}{\partial z} + \frac{\partial}{\partial z} \left[K(\theta) \cdot \frac{\partial \psi(\theta)}{\partial z} \right]$$

Richards' Equation

Highly nonlinear partial differential equation (PDE) that is solved numerically by specifying initial and boundary conditions, dividing soil up into thin layers, and iteratively solving for small time increments, such as in HYDRUS.

$$\frac{\partial \theta}{\partial t} = -\frac{\partial K(\theta)}{\partial z} + \frac{\partial}{\partial z} \left[K(\theta) \cdot \frac{\partial \psi(\theta)}{\partial z} \right]$$

Richards' Equation

Both $K(\theta)$ and $\psi(\theta)$ are required by the Richards equation. These can be estimated from soil-water retention curve parameters using the van Genuchten equations:

$$\theta(\psi) = \theta_r + \frac{\theta_s - \theta_r}{(1 + |\alpha\psi|^n)^m}$$

$$K(\psi) = K^* S_e^l \cdot [1 - (1 - S_e^{1/m})^m]^2$$

where n is the pore size distribution (>1), α is related to the inverse of the air entry pressure, $m = 1 - 1/n$, θ_s and θ_r are saturated and residual moisture content.

Issues with Applying Richards' Equation

- valid for small (1-meter) scale
- assumption of homogeneous soil profile
- highly nonlinear equation leads to numerical instability
- high computational cost due to the need to finely discretize (in space and time)



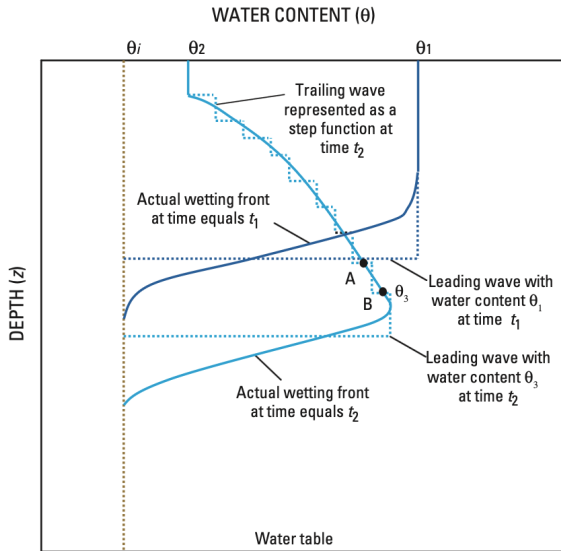
Kinematic Wave Approximation

Ignores capillary/pressure gradients and assumes unsaturated flow is only driven by gravity (1-D only).

$$\boxed{\frac{\partial \theta}{\partial t} = -\frac{\partial K(\theta)}{\partial z}} + \cancel{\frac{\partial}{\partial z} \left[K(\theta) \cdot \frac{\partial \phi(\theta)}{\partial z} \right]}$$

Far more computationally stable and efficient, but since negative pressure gradients are ignored, there will be an under-estimation of early time wetting front and infiltration rates. Cannot model lateral redistribution.

Kinematic Wave

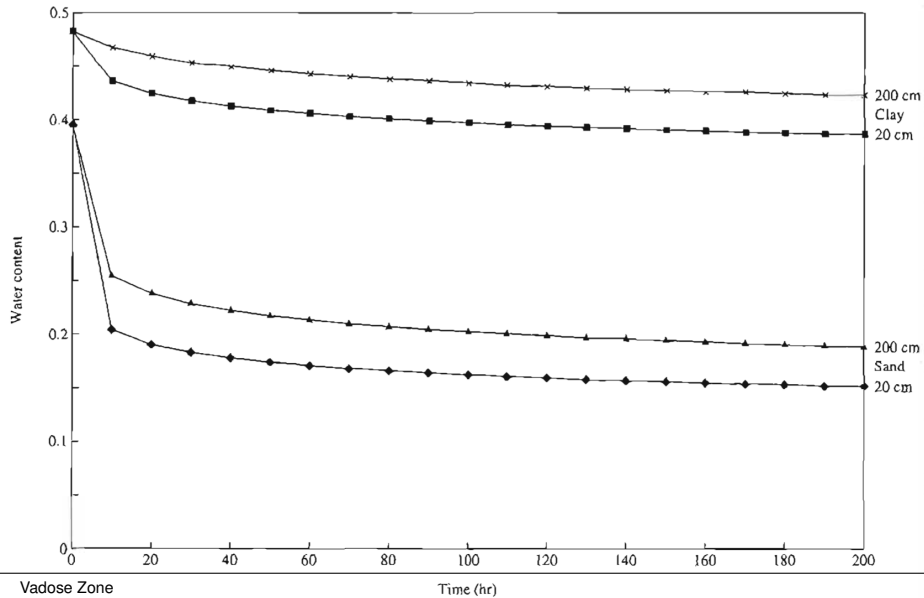


Redistribution

Following the water input event, water in the vadose zone is subject to **redistribution** by gravity and pressure forces and removed by evapotranspiration. This can be modeled using the Richards' equation with the addition of a "sink" term representing root water uptake.

$$\frac{\partial \theta}{\partial t} = -\frac{\partial K(\theta)}{\partial z} + \frac{\partial}{\partial z} \left[K(\theta) \cdot \frac{\partial \psi(\theta)}{\partial z} \right] - \mathbf{S}$$

Redistribution



Redistribution

